

Sindh Madressatul Islam University Karachi

Object Oriented Programming



SPRING 2026

Lecture 6 of 16
Engr. Majid Kaleem

WEEK 06 - AGENDA

- Inheritance in Java

<https://onecompiler.com/java>

WEEKLY AGENDA

TENTATIVE WEEKLY DATES		TENTATIVE TOPICS
1	26 th -Jan – 30 th Jan	INTRO. TO OBJECT-ORIENTED PROGRAMMING, PROCEDURAL VS.OOP, OOP PRINCIPLES, ETC.
2	02 nd Feb – 06 th Feb	CLASSES & OBJECTS, JAVA NAMING CONVENTION, JAVA PACKAGES, & THE FOUR PILLARS OF OOP
3	09 th Feb – 13 th Feb	METHODS & MODULAR PROGRAMMING
4	16 th Feb – 20 th Feb	ABSTRACT CLASSES & INTERFACES ASSIGNMENT #1 & QUIZ #1
5	23 rd Feb – 27 th Feb	USE OF STATIC KEYWORD IN JAVA
6	02 nd Mar – 06 th Mar	POLYMORPHISM
7	09 th Mar – 13 th Mar	INTERFACES ASSIGNMENT #2 & QUIZ #2
8	16 th Mar – 20 th Mar	EXCEPTION HANDLING
← MID TERM EXAMINATIONS (23rd MAR 26 – 27th MAR 2026) →		
9	30 th Mar – 03 rd Apr	COLLECTIONS FRAMEWORK
10	06 th Apr – 10 th Apr	FILE HANDLING
11	13 th Apr – 17 th Apr	GUI PROGRAMMING
12	20 th Apr – 24 th Apr	APPLICATION DEVELOPMENT ASSIGNMENT #3 & QUIZ #3
13	27 th Apr – 01 st May	DESIGN PRINCIPLES & PATTERNS
14	04 th May – 08 th May	ADVANCED OOP CONCEPTS
15	11 th May – 15 th May	CASE STUDIES AND INDUSTRY PRACTICES ASSIGNMENT #4 & QUIZ #4
16	18 th May – 22 nd May	PRESENTATIONS & REVISION WEEK
← FINAL TERM EXAMINATIONS (25th MAY 26 – 29th MAY 2026) →		

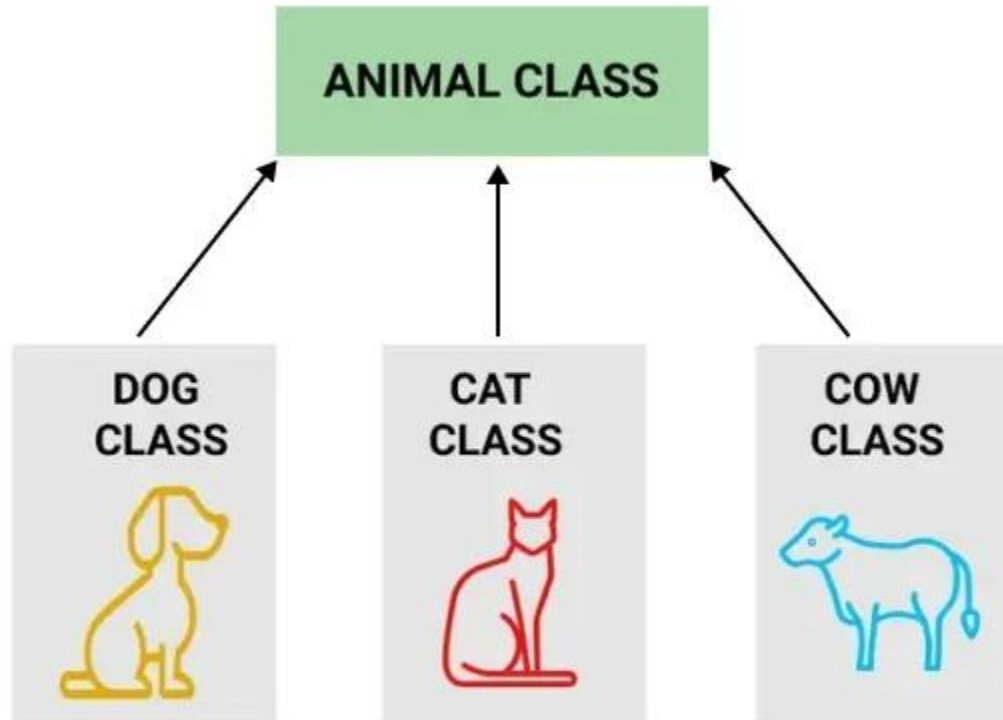
Note

- This schedule is subject to revisions as conditions may warrant.*
- Topics will be covered in sequence no matter if the university observes any planned or unplanned holidays.*
- The information in this course outline is subject to revision as conditions may warrant.*

INHERITANCE

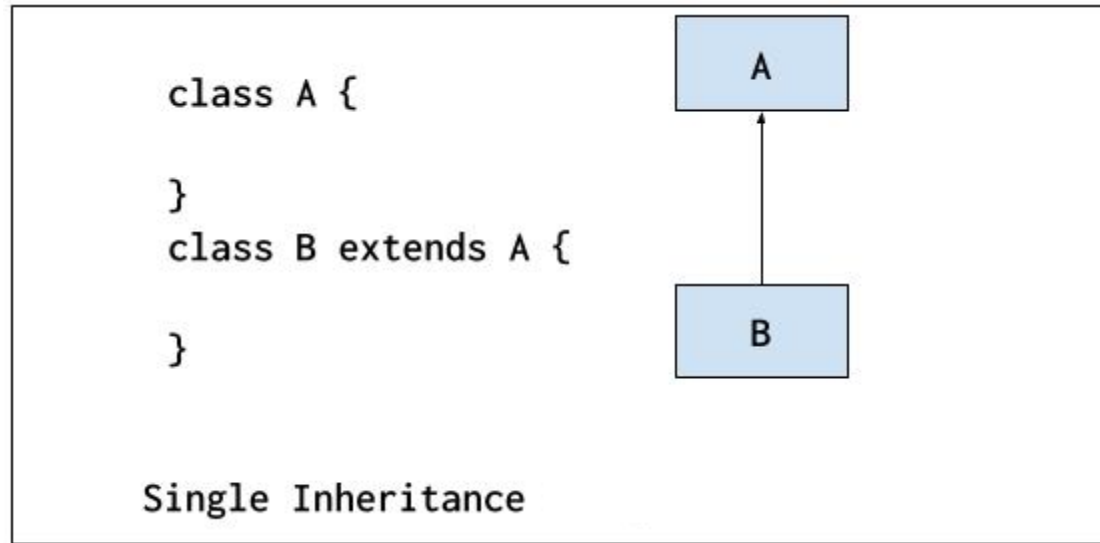
- All major uses of inheritance in Java:
 1. Single
 2. Multilevel
 3. Hierarchical
 4. Hybrid (via interfaces)
 5. Method overriding
 6. super keywords
 7. Abstract classes
 8. Interface inheritance
 9. Multiple inheritance via interfaces

INHERITANCE



INHERITANCE

- **Single Inheritance**

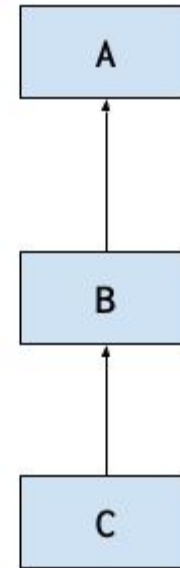


INHERITANCE

- **Multilevel Inheritance**

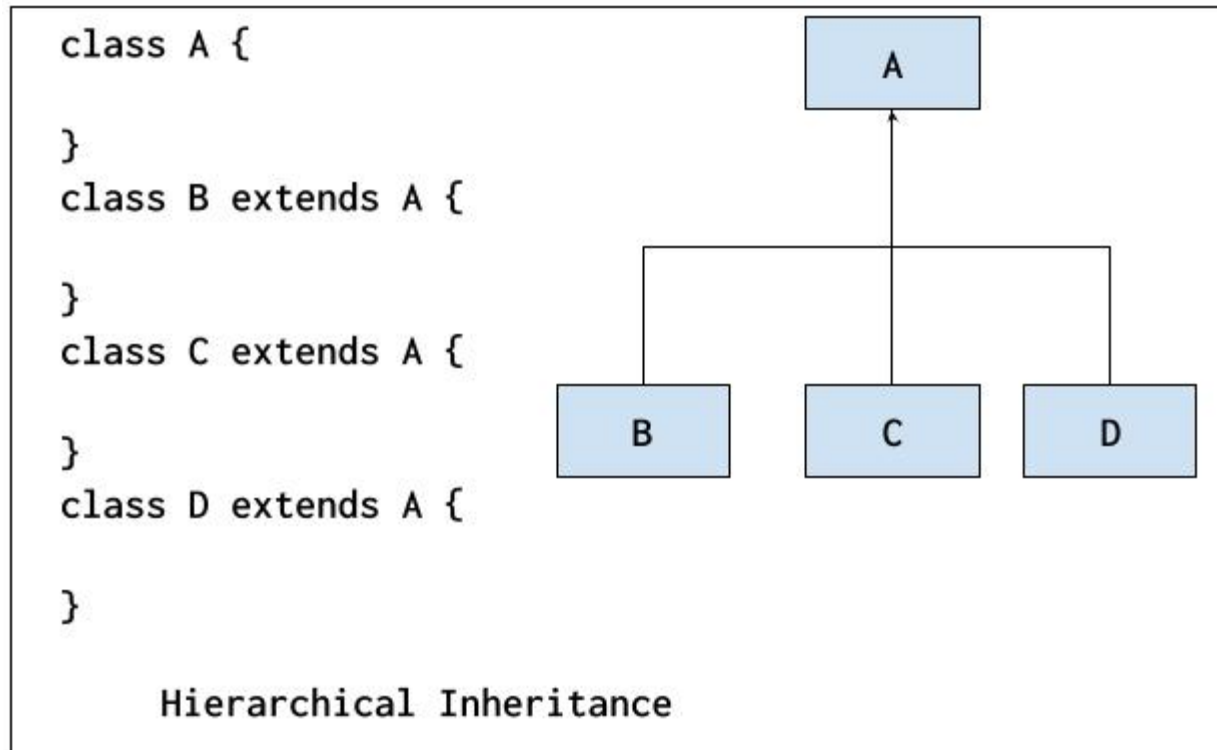
```
class A {  
  
}  
class B extends A {  
  
}  
class C extends B {  
  
}
```

Multilevel Inheritance



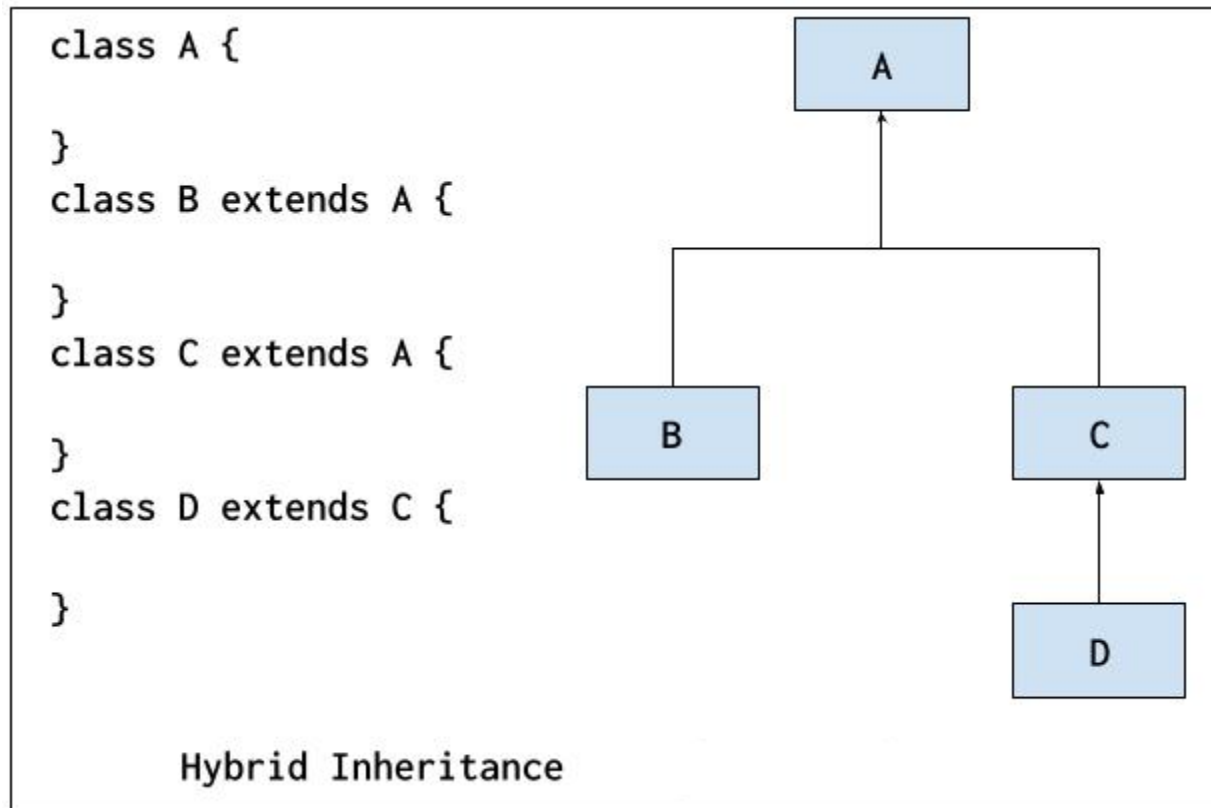
INHERITANCE

- **Hierarchical Inheritance**



INHERITANCE

- Hybrid Inheritance



To be continued...

```
If(anyQuestions)
{
    askNow();
}
else
{
    thankYou();
    submitAttendance();
    endClass();
}
```